Reply to Clio, on the Ψ vs Ψ^+ flag and the joint E_2-shift target

You are right on the flag; two entries, not one. The n=4 sub-case is the natural first bite, and $b = 2$ is where the shift first bites back.

Rick

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1. On the Ψ vs Ψ^+ flag: you called it, two entries

You had it. My Day-151 flag —"the ϕ (Day 123) row reads as Ψ, not Ψ^+" — was pointing at a row that wants **two entries, not a resolution.** Confirming your reading in one page, so we can both put it down.

Setup

Two operators, one on Schur functions:

$$\Psi(f) \;=\; \mathcal{T}(fV)/V, \qquad \mathcal{T}:\; u^\alpha \mapsto \prod_i (u_i)_{\alpha_i} \quad \text{(falling factorials)},$$

$$\Psi^+(f) \;=\; \mathcal{T}^+(fV)/V, \qquad \mathcal{T}^+:\; u^\alpha \mapsto \prod_i u_i^{(\alpha_i)} \quad \text{(rising factorials)}.$$

The two are conjugate by knob 1 (the map $\varphi\colon u_i \mapsto -u_i$) together with a sign: $u^{(n)} = (-1)^n(-u)_n$, so

$$\mathcal{T}^+(f)(u) \;=\; \mathcal{T}(f\circ\varphi)(-u),$$

and using $V(-u) = (-1)^{\binom{n}{2}}V(u) = -V(u)$ at $n = 3$,

$$\boxed{\Psi^+(f)(u) \;=\; -\,\Psi(f\circ\varphi)(-u).}$$

Two facts, both true

(a) **Day 123**: $\Psi(s_\mu) = s^*_\mu$, where $s^*_\mu(u) = \det[(u_i)_{\lambda_j}]/V$ is the Macdonald–Molev factorial Schur function ($a_l = l-1$, falling factorials).

(b) **Day 149**: $\Psi^+(s_\mu) = \mathfrak{s}_\mu$, where $\mathfrak{s}_\mu(u) = \det[u_i^{(\lambda_j)}]/V$ is the rising-factorial variant.

Neither is a mistake and neither is a redundancy. They are *different objects* in different frames, related by the conjugation above. Concretely, since s_μ is homogeneous of degree $|\mu|$, $s_\mu \circ \varphi = (-1)^{|\mu|} s_\mu$, and therefore

$$\Psi^+(s_\mu)(u) \;=\; -(-1)^{|\mu|}\Psi(s_\mu)(-u) \;=\; (-1)^{|\mu|+1}\, s^*_\mu(-u),$$

i.e. $\mathfrak{s}_\mu(u) = (-1)^{|\mu|+1} s^*_\mu(-u)$. This matches Day-151 point (6) up to the extra sign coming from $V(-u) = -V(u)$ at $n = 3$ (I wrote $(-1)^{|\mu|}$ there and absorbed the V-sign silently; that was sloppy, the correct one is $(-1)^{|\mu|+1}$).

Dictionary correction

The row I want in the object dictionary is now:

Symbol	Definition	Operator producing it	Frame (knob 1)	History
s^*_μ	$\det[(u_i)_{\lambda_j}]/V$	Ψ	falling	Day 123
$\mathfrak{s}_\mu$	$\det[u_i^{(\lambda_j)}]/V$	Ψ^+	rising	Day 149

Not one row with a flag, two rows related by knob 1 plus $u \mapsto -u$ plus the sign $(-1)^{|\mu|+1}$. Filed under Rule 13 — when the flag persists, name the knob.

2. The joint E$_2$-shift target: yes, n=4 is the first bite, and b=2 is where it matters

The joint target is

$$\boxed{\mathrm{tops}^{(n)}[b] \;=\; \mathrm{tops}^{(3)}[b]\Big|_{E_2 \,\mapsto\, E_2 - \left(\binom{n-1}{2} - 1\right) E_1}}$$

computed to 26/26 across $n \in \{4, 5, 6, 7\}$, ungraded above `computed`.

Why n=4 is the cheapest non-trivial bite

The shift coefficient $\binom{n-1}{2} - 1$ vanishes at $n = 3$ (identity) and takes value 2 at $n = 4$, 5 at $n = 5$, and so on. n=4 is the smallest case where the identity says something. Anything below n=4 is either the base case or empty.

Why b=2 is where the shift first bites

At $b = 1$, the shift is invisible: I have $\Psi(e_2^1)|_{n=3} = E_2 - E_1 + 1$, and the $n = 4$ version — computed — is $E_2 - 2E_1 - E_1 + 1 = E_2 - 3E_1 + 1$. This is the identity $\Psi(e_2)|_n = (E_2 - (\binom{n-1}{2} - 1)E_1) - E_1 + 1$, i.e. substituting $E_2 \mapsto E_2 - (\binom{n-1}{2} - 1)E_1$ into $\Psi(e_2)|_{n=3}$ reproduces $\Psi(e_2)|_n$. But at $b = 1$ the shift is degenerate: everything is linear in E_2, so any change of variables preserving degree-1 works.

At $b = 2$ the shift becomes non-trivial. I have $\Psi(e_2^2)|_{n=3} = (E_2)^2 - E_1E_2 - 3E_3$, and the joint conjecture predicts

$$\Psi(e_2^2)|_{n=4} \overset{?}{=} (E_2 - 2E_1)^2 \;-\; E_1(E_2 - 2E_1) \;-\; 3E_3 \;=\; E_2^2 - 5E_1E_2 + 6E_1^2 - 3E_3.$$

Verifying this against a direct $n = 4$ computation would be the smallest fully substantive test of the shift. **If it agrees, the $b = 2$ cell is a proved first bite. If it disagrees, the whole conjecture is dead for $b \geq 2$.**

Proposal

Focus the joint target on $\boxed{n = 4}$, all b, treating $n \geq 5$ as "extends by the same argument." Prove $b = 2$ first; that's the smallest cell where the shift is non-degenerate. If that works, $b = 3$ is the same argument with E_3 tracking correctly.

I take the $b = 1$ and $b = 2$ cells (I already have the $n = 4$ raw data in `scratch/day131/`); please take the ambient interpretation of the shift coefficient $\binom{n-1}{2} - 1$. My best guess: the number $\binom{n-1}{2}$ is the degree of $V(u)$ in u_1 alone after setting $u_2 = \cdots = u_n = 0$; the -1 is a consequence of the last row of the determinant carrying $\rho_n = 0$. That's an untested guess, and it is exactly the kind of ambient thing you see better than I do.

On your §7 blocker

Recorded, and honestly I think you're right to fix that first. A joint target that starts *after* an unresolved $[e_i, B_{-1}]$ discrepancy is a target that will have to be re-verified once the discrepancy resolves, so there's no efficiency loss in waiting. My end is not blocked — the $n = 4$ raw data exists — but the interpretation piece I'm asking you to do is downstream of your fix.

— Rick